Daniel Dela Dzikunu

Medical Student (MBChB) · AI Data & Language Work · Web Developer

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SUMMARY

First-year medical student in Accra who builds AI data and language tools. I designed a pipeline that turned my lecture slides into eight checked exam question banks (about 2,900 questions) that classmates revised from, found and fixed a defect in a published Twi speech dataset, and fine-tuned speech recognition models for Twi and Ewe. Native Twi (fluent reading and writing) and native-level English. Comfortable with careful, rule-bound review work and with writing up exactly how I reached a verdict. Available 15+ hours a week, remote, GMT.

Proof: huggingface.co/danieldzikunuofmarvel (dataset, models, live demo) • anatomy-qbank.vercel.app (one of the 8 banks) • dfourmarvel.github.io/flybrain-flappy

EDUCATION

Entrance University of Health Sciences 2026 – Present

Bachelor of Medicine and Bachelor of Surgery (MBChB) | Year 1 | Accra, Ghana

Google Cybersecurity Professional Certificate 2026 – Present

Google / Coursera, on an ITExperience scholarship | 5 of 8 courses complete | ISC2 Candidate (June 2026)

HEALTH EXPERIENCE

Blood Donation and Health Screening Days Jul 2026, Oct 2026

Volunteer, Entrance University of Health Sciences team | Spintex and Manet, Accra

- Went door to door before each event, answering residents' safety worries about donating blood.
- Ran finger-prick screening for blood group, hepatitis B, blood glucose, malaria and typhoid, one person after another for hours.

AI DATA AND LANGUAGE PROJECTS

Medical Exam Question Banks Oct 2026

8 live practice sites: anatomy, histology, physiology, genetics, medical sociology, organic chemistry, informatics, French

- Set the rules (slide-only, one defensible answer, an explanation for every answer). An AI drafted the items; a separate AI reviewer checked every answer key against the slides, and flagged items were rewritten.
- About 2,900 questions in five formats (single best answer, true/false, assertion-reason, matching, fill-in), each with an explanation.
- About 10 classmates revised from them for our October exams.

BibleTTS Twi Transcript Repair (published dataset) 2026

- Noticed that two Twi vowels (ɛ and ɔ) had been silently stripped from all ~21,600 transcripts of a public speech dataset, a defect only a Twi reader would catch.
- Rebuilt and published the corrected transcripts on Hugging Face (CC BY-SA 4.0), original and repaired text side by side, and reported it upstream on an issue unanswered since March 2024.

Twi and Ewe Speech Models 2026

- Fine-tuned Meta's MMS speech recognition model on Bible readings: Twi 17.2% word error rate (base model 27.6%), Ewe 6.9% (13.3%) on held-out test clips; I don't speak Ewe, so it was scored on the test set only. Also trained Twi and Ewe text-to-speech voices.
- Tested on everyday speech from new speakers, where the gain is small (Twi 50.9% vs 54.3%), and published that limit with the models and a live demo on Hugging Face.

Twi Bible Video Transcription 2024 – Present

- Transcribed Twi Bible readings from video with timestamps as early training data for my Twi speech project.
- Built a tool that turns Twi, Ewe and Ga Bible and hymn videos into an audio file plus the on-screen text in reading order, then checked its output against the audio.

SOFTWARE PROJECTS

med-match: specialty quiz for Ghanaian medical students 2026

- Web quiz that suggests specialties along the Ghana College of Physicians and Surgeons pathway, not US or UK match data; 21 completed assessments in its first 10 days (Sep 2026).
- Locked the database so browsers cannot read or write it directly; all saves go through the server.

Afahyɛ: event-vendor marketplace for Ghana 2026 – Present

- Building solo in Next.js and Supabase: vendor listings, customer briefs, two-sided job completion, saved vendors, admin tools, error monitoring and bot protection.
- Test each database security rule by signing in as real users of each role before it ships.

flybrain-flappy 2026

- Ran a simulation of a male fruit fly's fully mapped nervous-system wiring (Janelia/Google MaleCNS connectome) inside Flappy Bird through a fitted interface. The wiring relays the interface's signal; it does not learn the game.
- Pre-registered a 60-run test: real wiring scored a median 18.1 of a possible 22 on unseen levels, shuffled wiring 0 ($p \approx 7\times10^{-12}$). Playable browser version.

Two educational web games 2026

- Menstrual Health Awareness game separating facts from myths for Ghanaian players, and a Bible quiz with read-aloud that runs on low bandwidth.

CYBERSECURITY LAB WORK

- Malware triage of two flagged files with Detect It Easy and VirusTotal, ending in a written false-positive verdict.
- Wrote and tested Suricata detection rules; analysed captures in Wireshark and tcpdump; investigated a breach with SQL log queries; decrypted AES-256 files and verified integrity with SHA-256.
- Each lab written up as a formal report and a public post on my site.

LANGUAGES AND SKILLS

Languages: Twi (native; fluent reading and writing), English (native fluency; educated in English), Fante (conversational), French (intermediate)

AI and data: Dataset review and repair, transcription checking, fine-tuning speech models (Hugging Face), designing AI review pipelines, Python

Web: Next.js, TypeScript, Supabase (PostgreSQL), HTML, CSS, JavaScript, Git and GitHub

Security: Linux command line, Wireshark, tcpdump, Suricata, SQL, OpenSSL, hash checks, malware triage